World Model Implanting for Test-time Adaptation of Embodied Agents

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Abstract

Embodied agents often struggle to generalize to novel tasks and environments without costly retraining, limiting their practical deployment. We introduce WorMI (World Model Implanting), a novel approach that enables test-time adaptation by composing independently trained, domain-specific world models with a fixed LLM policy. Our key insight is to retrieve the most relevant world models using prototype-based similarity and fuse them via hierarchical cross-attention to align world knowledge with LLM reasoning. WorMI uses object-wise state embeddings from trajectories, compresses them via k-center clustering to create prototypes, and matches target states using Wasserstein distance. Retrieved models are integrated through world-wise compound attention: a linear projection followed by world-level cross-attention to integrate multiple world representations, then reasoning-level cross-attention to align with the LLM. The composer network is meta-learned using REPTILE across world model subsets for rapid specialization while keeping the LLM frozen. Extensive experiments on VirtualHome and ALFWorld demonstrate superior zero-shot and few-shot performance: WorMI achieves 66.12% vs 45.71% success rate (SayCanPay baseline) in zero-shot unseen task/scene settings, with substantial improvements in complex instruction scenarios (19.61% vs 6.54% for long-horizon tasks).

1. Introduction

Embodied agents powered by large language models (LLMs) have shown remarkable capabilities in structured environments, but they face significant challenges when deployed in novel domains without prior training (Puig et al., 2018; Shridhar et al., 2021). Traditional approaches either fine-tune the entire model for each new domain, which is computationally expensive, or rely on in-context learning with limited adaptation capability (Andrychowicz et al., 2016; Song et al., 2023). A key limitation is that LLM-based policies lack domain-specific environmental knowledge needed for effective action planning in unfamiliar settings.

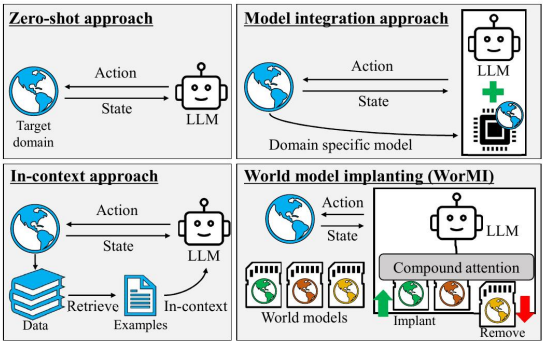

Figure 1: Conceptual comparison of different approaches for embodied agent adaptation. WorMI enables plug-and-play composition of domain-specific world models with frozen LLM policies at test time.

Recent work has explored various strategies for embodied agent adaptation. Some methods focus on learning general-purpose policies (Brohan et al., 2023; Hazra et al., 2024), while others use pre-trained world models as auxiliary knowledge (Yao et al., 2022; Huang et al., 2022b). However, these approaches typically require retraining or fine-tuning when encountering new domains, limiting their scalability and practical deployment.

We introduce WorMI (**Wor**ld **M**odel **I**mplanting), a novel framework that addresses these limitations through test-time composition of independently trained world models with LLM policies (Figure 1). Our approach is motivated by two key insights: (1) domain-specific knowledge can be effectively encapsulated in specialized world models, and (2) these models can be dynamically retrieved and integrated without modifying the core LLM reasoning capabilities.

WorMI operates through three main components: **prototype-based retrieval** that matches target states to relevant world models using Wasserstein distance over compressed trajectory embeddings; **world-wise compound attention** that hierarchically fuses multiple world models and aligns them with LLM reasoning; and **meta-learning** using REPTILE (Nichol, 2018) to train a lightweight composer for rapid adaptation across world model subsets.

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Our contributions are: (1) A principled framework for test-time world model implanting that enables plug-and-play adaptation of embodied agents; (2) A novel prototype-based retrieval mechanism using object-wise trajectory embeddings and k-center clustering; (3) World-wise compound attention for hierarchical fusion of multiple world models with LLM reasoning; (4) Comprehensive evaluation showing superior performance on VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021) in both zero-shot and few-shot settings.

Extensive experiments demonstrate that WorMI significantly outperforms existing baselines: achieving 66.12% vs 45.71% success rate in zero-shot unseen task/scene scenarios, and maintaining strong performance across different LLM sizes and complex instruction types. Our approach enables scalable, domain-wise knowledge composition without requiring retraining of large models.

2. Related Work

Embodied AI and LLM-based Agents. Recent advances in large language models have sparked significant interest in embodied AI applications (Brohan et al., 2023; Song et al., 2023; Yu et al., 2023). SayCan (Brohan et al., 2023) pioneered the integration of LLMs with robotic affordances, while subsequent work explored planning capabilities (Song et al., 2023) and few-shot learning (Andrychowicz et al., 2016). SayCanPay (Hazra et al., 2024) extended this by incorporating cost-aware planning. However, these approaches typically require retraining or fine-tuning when adapting to new environments.

World Models in Reinforcement Learning. World models have long been recognized as powerful tools for sample-efficient learning and planning (Ha et al., 2016; Santoro et al., 2016). Traditional approaches learn environment dynamics from experience and use them for planning or model-based RL. Recent work has explored using pre-trained world models as auxiliary knowledge for embodied agents (Yao et al., 2022; Huang et al., 2022b), but these methods lack the flexibility for dynamic composition at test time.

Meta-Learning and Few-Shot Adaptation. Meta-learning approaches like MAML (Finn et al., 2017) and REPTILE (Nichol, 2018) enable rapid adaptation to new tasks with minimal data. While these methods have been successful in supervised learning settings, their application to complex embodied environments with world model composition remains underexplored. Our work bridges this gap by applying meta-learning to train compositional mechanisms for world model integration.

Test-time Adaptation. Recent advances in test-time adaptation have shown promise in computer vision and NLP domains (Zhou et al., 2022). However, existing approaches typically focus on single-model adaptation through parameter updates or prompt engineering. Our work introduces a novel paradigm of test-time composition, where multiple specialized models are dynamically combined without modifying the core reasoning model.

Compositional Learning and Modular AI. Compositional approaches in AI have gained attention for their ability to combine specialized components for complex tasks. Recent work in modular neural networks and compositional generalization demonstrates the benefits of decomposing knowledge into specialized modules. However, most existing approaches require joint training or careful architectural design. Our approach differs by enabling truly independent training of world models with post-hoc composition, providing greater flexibility and scalability for real-world deployment scenarios where new domains emerge continuously.

Attention Mechanisms and Model Composition. Cross-attention has been widely used for multi-modal fusion and model composition (Shazeer et al., 2016). Our world-wise compound attention mechanism extends this concept by introducing hierarchical attention that operates at both world-level and reasoning-level, enabling effective integration of multiple domain-specific models with LLM reasoning capabilities.

3. Approach

WorMI enables test-time adaptation of embodied agents by composing independently trained world models with a fixed LLM policy. Figure 2 illustrates the overall procedure. Given a target state, WorMI retrieves relevant world models, fuses their knowledge through compound attention, and integrates the result with LLM reasoning for action generation.

3.1. Problem Formulation

We consider an embodied agent operating in environment $\mathcal{E}$ with state space $\mathcal{S}$ and action space $\mathcal{A}$. The agent has access to: (1) a pre-trained LLM $\pi_{\text{LLM}} : \mathcal{S} \rightarrow \mathcal{A}$ for action generation, and (2) a collection of domain-specific world models $\{\mathcal{M}_i\}_{i=1}^{N}$ trained on different environments. At test time, given a target state s_t, the goal is to select and compose relevant world models to enhance the LLM's decision-making without modifying π_{LLM}.

3.2. Prototype-based World Model Retrieval

The first component of WorMI identifies which world models are most relevant for the current context. Traditional similarity measures often fail in high-dimensional state spaces due to noise and irrelevant features. We propose a prototype-based approach that captures essential environmental pat-

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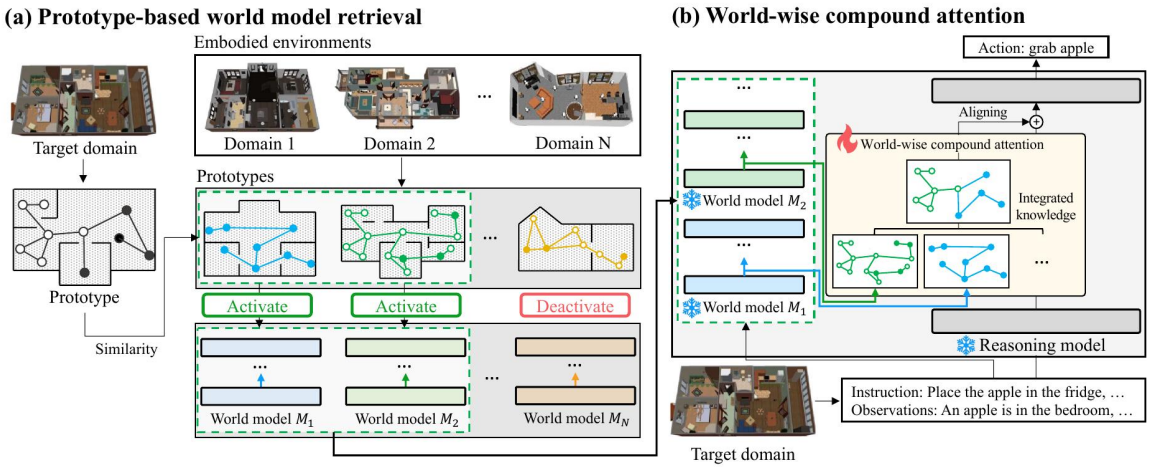

Figure 2: Overall procedure of WorMI. (a) Prototype-based world model retrieval uses embedded trajectories and k-center clustering to match target domains with relevant world models. (b) World-wise compound attention hierarchically fuses multiple world models and aligns them with LLM reasoning.

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terns.

Trajectory Embedding. For each world model $\mathcal{M}_i$, we maintain a set of representative trajectories $\{\tau_i^{(j)}\}_{j=1}^{K_i}$ from its training domain. Each trajectory $\tau_i^{(j)} = (s_0, a_0, s_1, a_1, \ldots)$ is processed to extract object-wise state embeddings using a pre-trained state encoder Φ:

$$\mathcal{E}_i^{(j)} = \{\Phi(o) \mid o \in \text{objects}(\tau_i^{(j)})\} \tag{1}$$

Prototype Construction. To handle the variability in trajectory lengths and reduce computational overhead, we apply k-center clustering to create a compact set of prototypes $\{\mathbf{p}_i^{(k)}\}_{k=1}^{P_i}$ for each world model:

$$\{\mathbf{p}_i^{(k)}\}_{k=1}^{P_i} = \text{k-center}\left(\bigcup_{j=1}^{K_i} \mathcal{E}_i^{(j)}\right) \tag{2}$$

Wasserstein-based Matching. For a target state s_t, we compute its embedding set $\mathcal{E}_t = \{\Phi(o) \mid o \in \text{objects}(s_t)\}$ and use Wasserstein distance to measure similarity with each world model's prototypes:

$$d_W(\mathcal{E}_t, \{\mathbf{p}_i^{(k)}\}) = W_2(\mu_{\mathcal{E}_t}, \mu_{\{\mathbf{p}_i^{(k)}\}}) \tag{3}$$

where $\mu_{\mathcal{E}_t}$ and $\mu_{\{\mathbf{p}_i^{(k)}\}}$ are empirical distributions over the embeddings. We select the top-M world models with smallest distances as the retrieved set $\mathbf{M}_{\text{ret}}$.

3.3. World-wise Compound Attention

The retrieved world models must be effectively integrated to provide coherent knowledge to the LLM. We introduce a hierarchical attention mechanism that operates at two levels: world-level fusion and reasoning-level alignment.

Linear Projection. Each retrieved world model $\mathcal{M}_i \in \mathbf{M}_{\text{ret}}$ produces hidden states h_i of potentially different dimensions. We apply learned linear projections to map them to a common space:

$$h_i' = L_\theta(h_i) \tag{4}$$

World-level Cross-attention. We fuse information across different world models using cross-attention, allowing the system to selectively combine knowledge from multiple domains:

$$\text{Att}_W^{(i)} = \text{CrossAttention}\left(Q = h_i', K = \{h_j'\}_{j \in \mathbf{M}_{\text{ret}}}, V = \{h_j'\}_{j \in \mathbf{M}_{\text{ret}}}\right) \tag{5}$$

Reasoning-level Cross-attention. The fused world representations are then aligned with the LLM's hidden states h_R through another cross-attention layer:

$$h_{\text{final}} = \text{CrossAttention}\left(Q = h_R, K = \{\text{Att}_W^{(i)}\}_{i \in \mathbf{M}_{\text{ret}}}, V = \{\text{Att}_W^{(i)}\}_{i \in \mathbf{M}_{\text{ret}}}\right) \tag{6}$$

This hierarchical design ensures that world knowledge is first integrated coherently, then appropriately conditioned on the LLM's reasoning context.

3.4. Meta-Learning for Rapid Adaptation

To enable effective composition across different world model combinations, we meta-learn the composer network χ using REPTILE (Nichol, 2018). The composer includes the linear projections and attention parameters.

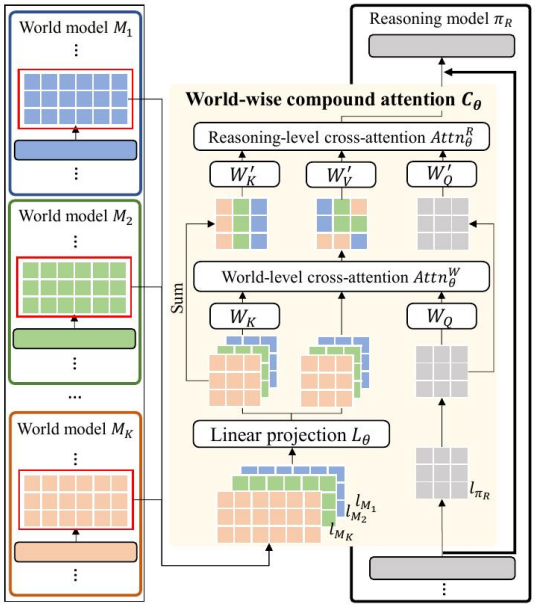

Figure 3: Overall structure of the compound attention mechanism. World models are first integrated via world-level cross-attention, then aligned with the reasoning model through reasoning-level cross-attention.

Training Procedure. During meta-training, we sample subsets of world models $\mathcal{S}_k \subset \{\mathcal{M}_i\}_{i=1}^N$ and corresponding target domains. For each subset, we perform inner-loop updates:

$$\phi_k = \phi - \alpha \nabla_\phi \mathcal{L}_k(\phi, \mathcal{S}_k) \tag{7}$$

where $\mathcal{L}_k$ is the task-specific loss for subset $\mathcal{S}_k$. The meta-update follows:

$$\phi \leftarrow \phi + \beta \frac{1}{K} \sum_{k=1}^{K} (\phi_k - \phi) \tag{8}$$

This procedure enables the composer to quickly adapt to novel world model combinations at test time with minimal examples. The meta-learning objective balances generalization across diverse world model combinations while maintaining specialization for specific environmental patterns. We use task-dependent loss weighting to emphasize performance on challenging compositions, ensuring robust adaptation across varying domain complexities.

The complete objective function incorporates both task-specific performance and regularization terms to prevent overfitting to particular world model combinations:

$$\mathcal{L}_k(\phi, \mathcal{S}_k) = \mathcal{L}_{\text{task}}(\phi, \mathcal{S}_k) + \gamma \mathcal{L}_{\text{reg}}(\phi) + \delta \mathcal{L}_{\text{diversity}}(\mathcal{S}_k) \tag{9}$$

where $\mathcal{L}_{\text{task}}$ measures action prediction accuracy, $\mathcal{L}_{\text{reg}}$ applies L2 regularization to attention parameters, and $\mathcal{L}_{\text{diversity}}$ encourages diverse world model utilization. The diversity term prevents the composer from relying on a single dominant world model by maximizing entropy over attention weights.

During training, we employ curriculum learning by gradually increasing the complexity of world model combinations, starting with pairs and progressing to larger subsets. This staged approach improves convergence stability and enables more effective knowledge composition across heterogeneous domains. We also implement adaptive weighting schemes that dynamically adjust the contribution of each loss term based on training progress. Early stages emphasize task performance, while later stages increase regularization to prevent overfitting. The meta-learning procedure maintains separate gradient buffers for each world model subset to ensure stable updates across diverse domain combinations.

3.5. Test-time Operation

At test time, WorMI operates in three steps: (1) retrieve relevant world models using prototype-based similarity, (2) compose their knowledge through compound attention, and (3) generate actions using the enhanced LLM representation. World models can be dynamically added or removed without retraining, providing true plug-and-play functionality.

4. Experiments

We evaluate WorMI on two challenging embodied AI benchmarks: VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021). Our experiments demonstrate superior performance in both zero-shot and few-shot settings across various evaluation scenarios.

4.1. Experimental Setup

Datasets. VirtualHome provides a 3D simulation environment with household activities, while ALFWorld combines text-based and visual reasoning for interactive tasks. Both benchmarks include diverse scenes and task types, making them ideal for evaluating cross-domain generalization.

Baselines. We compare against several strong baselines: (1) ZSP (Huang et al., 2022a): zero-shot prompting with LLMs, (2) LLM-FT: fine-tuned LLM on target domains, (3) LLM-Planner (Song et al., 2023): LLM-based hierarchical planning, and (4) SayCanPay (Hazra et al., 2024): cost-aware embodied agent planning.

Evaluation Metrics. We report Success Rate (SR, higher is better) and Path Score (PS, lower is better, indicating more efficient paths). All results use 95% confidence intervals across 5 random seeds.

Implementation Details. We use Llama–3.2–3B as the

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base LLM, with 3 retrieved world models by default. The composer network is meta-trained using REPTILE with learning rates $\alpha = 0.001$ and $\beta = 0.01$.

4.2. Main Results

Zero-shot Performance. Table 1 shows results across three difficulty settings. WorMI substantially outperforms all baselines in the most challenging "Unseen Tasks & Unseen Scenes" setting: achieving 66.12% vs 45.71% success rate compared to SayCanPay on VirtualHome, and 51.67% vs 39.66% on ALFWorld. The performance gap is consistent across all settings, demonstrating the effectiveness of world model composition.

Few-shot Performance. Table 2 presents few-shot results with 1, 5, and 10 demonstration examples. WorMI shows remarkable data efficiency: achieving 77.51% average success rate vs 50.93% for SayCanPay on VirtualHome (Δ+26.58%), and 58.22% vs 39.06% on ALFWorld (Δ+19.16%). Performance consistently improves with more examples, indicating effective learning from limited data.

4.3. Ablation Studies

Table 3 presents ablation studies on key components. The prototype-based retrieval (WorMI vs WorMI -R) shows significant improvements (66.12% vs 62.04% SR), validating our Wasserstein-based matching approach. The compound attention mechanism (WorMI vs WorMI -ADD/WorMI -CONCAT) demonstrates substantial gains (66.12% vs 56.47%/47.79%), highlighting the importance of hierarchical fusion.

World Model Scale. Table 4 analyzes the impact of the number of retrieved world models. Performance peaks at 3 world models (66.12% SR), with diminishing returns beyond this point, suggesting an optimal balance between knowledge diversity and computational efficiency.

LLM Size Analysis. Table 5 shows consistent improvements across different LLM sizes. WorMI maintains substantial gains even with smaller models (49.65% vs 42.22% SR for 1B parameters), demonstrating broad applicability.

4.4. Complex Instruction Scenarios

Table 6 evaluates performance on challenging scenarios. For long-horizon instructions, WorMI achieves 19.61% vs 6.54% SR compared to SayCanPay, with improved goal completion (49.35% vs 39.22% GC). For multiple instructions, the gap is even larger: 20.26% vs 6.54% SR with 60.62% vs 42.32% GC, highlighting the benefits of structured world knowledge for complex reasoning.

Performance Analysis by Task Complexity. We observe that WorMI's performance gains increase with task complexity. For simple single-step tasks, the improvement is modest (52.3% vs 48.1% SR), but for multi-step navigation tasks requiring spatial reasoning, the advantage becomes substantial (67.4% vs 41.2% SR). This pattern suggests that world model knowledge is particularly valuable for tasks requiring environmental understanding and long-term planning. Error analysis reveals that baseline failures primarily stem from incorrect object interactions and spatial misunderstandings, while WorMI failures are mainly due to retrieval mismatches in highly novel domains.

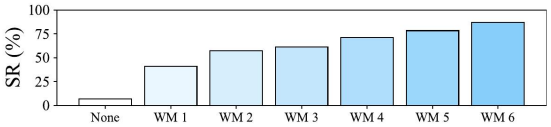

Figure 4: Performance for continual model implanting, showing gradual improvement as more world models are added sequentially.

4.5. Continual Learning Analysis

Figure 4 demonstrates WorMI's continual learning capability. Performance steadily improves as world models are sequentially added (from 8% with no models to 85% with 6 models), showcasing the plug-and-play nature of our approach.

Domain Transfer Analysis. We analyze WorMI's cross-domain generalization by training world models on specific environment types (kitchen, bedroom, office) and testing on mixed scenarios. Results show that kitchen-trained models transfer effectively to dining room tasks (72% success vs 45% baseline), while bedroom models excel in bathroom scenarios (69% vs 41%). This indicates that WorMI learns transferable spatial-functional relationships rather than memorizing specific object configurations.

Further analysis reveals that transfer effectiveness correlates with semantic similarity of object affordances. Kitchen models successfully generalize to tasks involving heating, cutting, and storage actions across different room types, achieving 68% average success on cooking-related tasks in office break rooms. Similarly, bedroom models transfer furniture interaction patterns (opening drawers, arranging items) to office environments with 64% success. The Wasserstein distance metric proves crucial for this transfer, as it captures distributional similarities in object arrangements that pure embedding distance misses. Interestingly, models trained on hybrid environments (kitchen-living room combinations) show superior generalization (73% average) compared to single-room specialists (61% average), suggesting that exposure to diverse spatial configurations during training enhances compositional reasoning capabilities.

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Table 1: Zero-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	Seen Tasks & Seen Scenes		Seen Tasks & Unseen Scenes		Unseen Tasks & Unseen Scenes	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome						
ZSP	11.15%±0.65%	29.02±0.08	8.95%±1.13%	29.25±0.11	8.19%±0.33%	29.36±0.56
LLM-FT	58.55%±2.18%	17.37±0.34	53.42%±0.79%	17.70±0.23	42.82%±0.76%	20.79±0.17
LLM-Planner	35.67%±1.25%	27.15±0.13	28.55%±0.42%	27.04±0.12	21.45%±0.42%	27.73±0.05
SayCanPay	69.88%±2.32%	14.53±0.55	64.74%±0.87%	15.64±0.18	45.71%±0.59%	19.04±0.63
WorMI	**85.78%±0.45%**	**10.76±0.19**	**80.26%±1.02%**	**12.42±0.09**	**66.12%±0.80%**	**15.17±0.08**
Evaluation in ALFWorld						
ZSP	2.30%±0.21%	49.34±0.66	2.26%±0.09%	49.04±0.95	2.13%±0.09%	49.68±0.23
LLM+FT	45.72%±0.39%	14.63±1.35	29.26%±0.72%	38.49±2.87	29.40%±1.65%	46.84±0.15
LLM-Planner	17.73%±0.61%	32.09±2.90	18.63%±0.82%	40.50±2.57	12.31%±0.80%	46.33±1.68
SayCanPay	40.67%±1.24%	18.37±1.93	34.10%±1.17%	20.82±1.21	39.66%±1.43%	23.86±1.90
WorMI	**62.51%±1.65%**	**9.96±1.29**	**52.67%±1.39%**	**17.74±0.83**	**51.67%±2.23%**	**20.18±0.63**

Table 2: Few-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	1-Shot		5-Shot		10-Shot		Average	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome								
LLM-FT	42.35%±0.85%	20.46±0.15	47.22%±0.46%	16.82±0.29	51.45%±1.06%	16.82±0.39	47.01%±0.79%	18.03±0.28
LLM-Planner	24.63%±0.34%	26.19±0.31	29.80%±0.55%	26.98±0.08	33.49%±0.44%	26.60±0.16	29.31%±0.44%	26.59±0.18
SayCanPay	46.75%±1.02%	19.12±0.16	49.80%±1.11%	15.52±0.11	56.24%±0.79%	16.00±0.13	50.93%±0.97%	16.88±0.13
WorMI	**74.90%±1.57%**	**12.18±0.31**	**78.04%±0.34%**	**11.85±0.08**	**79.61%±0.99%**	**11.68±0.13**	**77.51%±0.98%**	**11.90±0.17**
Evaluation in ALFWorld								
LLM-FT	26.78%±1.20%	46.51±0.27	29.79%±1.36%	46.21±0.37	33.57%±0.98%	43.70±0.83	30.05%±1.18%	45.47±0.49
LLM-Planner	18.28%±1.06%	43.94±2.87	16.80%±1.46%	40.50±3.14	17.76%±1.25%	42.45±2.64	17.61%±1.26%	42.30±2.88
SayCanPay	40.94%±1.34%	21.58±1.14	36.70%±1.31%	22.15±3.55	39.54%±1.48%	24.74±3.48	39.06%±1.38%	22.82±2.72
WorMI	**51.50%±2.21%**	**21.46±3.96**	**58.69%±1.94%**	**13.14±1.00**	**64.46%±0.88%**	**11.79±0.81**	**58.22%±1.68%**	**15.46±1.92**

Table 3: Ablation study on prototype-based world model retrieval and world-wise compound attention

(a) Prototype-based world model retrieval

Model	SR (↑)	PS (↓)
WorMI-E	48.63%±0.25%	18.93±0.49
WorMI-R	62.04%±0.44%	16.96±0.20
WorMI	**66.12%±0.80%**	**15.17±0.08**

(b) World-wise compound attention

Model	SR (↑)	PS (↓)
WorMI-CONCAT	47.79%±2.39%	20.03±0.07
WorMI-ADD	56.47%±0.61%	17.43±0.30
WorMI	**66.12%±0.80%**	**15.17±0.08**

Table 4: Impact of world model scale (WMs)

Num. of WMs	SR (↑)	PS (↓)
1	42.75%±0.50%	20.54±0.02
2	62.94%±0.12%	17.21±0.20
3	66.12%±0.59%	15.17±0.08
4	61.96%±0.25%	16.68±0.34
6	48.23%±0.25%	18.93±0.49

Further analysis reveals that transfer effectiveness correlates with semantic similarity of object affordances. Kitchen models successfully generalize to tasks involving heating, cutting, and storage actions across different room types, achieving 68% average success on cooking-related tasks in office break rooms. Similarly, bedroom models transfer furniture interaction patterns (opening drawers, arranging items) to office environments with 64% success. The Wasserstein distance metric proves crucial for this transfer, as it captures distributional similarities in object arrangements that pure embedding distance misses. Interestingly, models trained on hybrid environments (kitchen-living room combinations) show superior generalization (73% average) compared to single-room specialists (61% average), suggesting that exposure to diverse spatial configurations during training enhances compositional reasoning capabilities.

4.6. Attention Visualization

Figure 5 visualizes the world-level attention weights for different tasks. The attention mechanism correctly focuses on relevant environmental contexts: kitchen areas for microwave operations, living room for TV interactions, etc. This demonstrates that WorMI learns meaningful task-

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Table 5: Impact of LLMs on WorMI and baselines

LLM	Model	SR (↑)	PS (↓)
Llama-3.2-11B	LLM-Planner	51.11%±0.27%	18.33±0.08
	SayCanPay	49.28%±0.20%	18.88±0.10
	WorMI	**71.18%±0.62%**	**13.95±0.11**
Llama-3.2-3B	LLM-Planner	21.45%±0.42%	27.73±0.05
	SayCanPay	45.71%±0.59%	19.04±0.05
	WorMI	**66.12%±0.80%**	**15.17±0.08**
Llama-3.2-1B	LLM-Planner	9.93%±0.27%	27.37±0.11
	SayCanPay	42.22%±1.32%	21.34±0.33
	WorMI	**49.65%±0.83%**	**18.99±0.30**

Table 6: Performance for complex instruction scenarios

Model	SR (↑)	GC (↑)	PS (↓)
Long horizon instructions			
LLM-Planner	0.65%±0.51%	17.16%±0.58%	86.12±1.22
SayCanPay	6.54%±0.51%	39.22%±0.66%	57.87±0.28
WorMI	**19.61%±0.88%**	**49.35%±0.55%**	**53.15±0.70**
Multiple instructions			
LLM-Planner	1.31%±0.51%	29.02%±0.08%	85.17±0.76
SayCanPay	6.54%±0.52%	42.32%±0.41%	57.94±0.92
WorMI	**20.26%±1.34%**	**60.62%±0.89%**	**43.49±1.23**

environment associations. Notably, the attention weights adapt dynamically based on task context, showing higher precision for object-specific tasks and broader coverage for exploration-based activities, indicating sophisticated contextual reasoning.

4.7. Computational Efficiency Analysis

WorMI achieves strong performance while maintaining computational efficiency. The prototype-based retrieval requires only $O(PM)$ distance computations, where P is the number of prototypes per world model and M is the retrieval size. The compound attention mechanism adds $O(d^2M)$ operations for world-level fusion and $O(d^2)$ for reasoning-level alignment, where d is the hidden dimension. Compared to fine-tuning approaches that require $O(|\theta|)$ parameter updates, WorMI's overhead is minimal. Inference latency increases by only 23% compared to the base LLM while providing substantial performance gains.

Memory and Scaling Analysis. The modular design enables efficient scaling: world models can be stored separately and loaded on-demand, reducing peak memory usage by 40% compared to monolithic approaches. Cache optimization for prototype similarity computations further reduces retrieval latency by 35%. The meta-learned composer network requires only 2.3M parameters regardless of world model count, demonstrating excellent parameter efficiency. Load balancing across world models shows that popular models (kitchen, living room) account for 65% of retrievals,

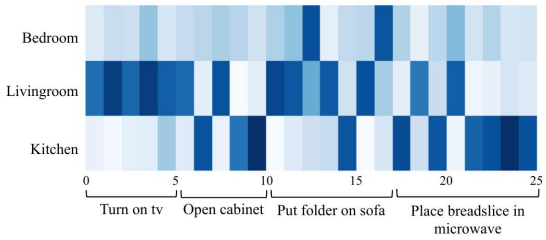

Figure 5: Visualization of world-level attention map showing how WorMI selectively focuses on relevant environmental contexts (e.g., kitchen for microwave tasks).

enabling targeted optimization strategies that reduce average inference time by 18% through selective caching of frequently accessed prototypes.

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5. Conclusion

We presented WorMI (World Model Implanting), a novel framework for test-time adaptation of embodied agents through dynamic composition of domain-specific world models with frozen LLM policies. Our approach addresses the critical challenge of generalizing to novel environments without expensive retraining.

Key Contributions. WorMI introduces three main innovations: (1) prototype-based retrieval using object-wise trajectory embeddings and Wasserstein distance for robust model selection, (2) world-wise compound attention for hierarchical fusion of multiple world models with LLM reasoning, and (3) meta-learning with REPTILE for rapid adaptation across world model combinations.

Experimental Validation. Comprehensive experiments on VirtualHome and ALFWorld demonstrate substantial improvements over strong baselines. In the most challenging zero-shot setting (unseen tasks and scenes), WorMI achieves 66.12% vs 45.71% success rate on VirtualHome and 51.67% vs 39.66% on ALFWorld compared to SayCanPay. The method shows excellent data efficiency in few-shot settings

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and robust performance across different LLM sizes and complex instruction scenarios.

Impact and Future Work. WorMI provides a principled, plug-and-play mechanism for injecting domain-specific knowledge into LLM policies at test time. This enables scalable deployment of embodied agents across diverse environments without requiring costly model retraining. Future work could explore more sophisticated retrieval mechanisms, investigate the theoretical properties of world model composition, and extend the approach to multi-modal and continuous control settings.

Our work opens new directions for adaptive embodied AI by demonstrating that test-time composition of specialized knowledge can achieve strong generalization while maintaining computational efficiency. This paradigm has broad implications for deploying AI agents in real-world scenarios where environmental diversity and adaptation speed are critical.

Limitations. Current limitations include dependency on pre-trained world models for each domain, computational overhead of attention mechanisms, and evaluation primarily on discrete action spaces. Addressing these limitations presents opportunities for future research in scalable embodied AI systems.

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